Questions for 1st Midterm Exam ECE 330:

Name: _____

Question #1 (6 points total):

The following differential equation is given:
$\frac{dy}{dt} = y^2 + 2yt + t^2$
Derive explicitly the update equation for y using the Improved Euler Method.
(6 points)

a) $\frac{dy}{dt} = y^2 + 2yt + t^2 = f(y,t)$

Update equation for Improved Euler Method,

$y(t+h) = y(t) + \frac{h}{2}\left(k^{(1)} + k^{(2)}\right)$

with $k^{(1)} = f(y,t)$

$k^{(2)} = f(y+hk^{(1)}, t+h)$

$\Rightarrow k^{(1)} = f(y,t) = y^2 + 2yt + t^2$

$k^{(2)} = f(y+hk^{(1)}, t+h) = (y+hk^{(1)})^2 + 2(y+hk^{(1)})(t+h) + (t+h)^2$

$\Rightarrow y(t+h) = y(t) + \frac{h}{2}\left(k^{(1)} + k^{(2)}\right)$

Question #2 (6 points total):

Derive explicitly the normal equations for a best fit parabola: $P(x) = a_0 + a_1 x + a_2 x^2$ to M measured data points (x_i, y_i) using the Least Squares Fitting Method.
(6 points)

Quadratic error:

$$F(a_0, a_1, a_2) = \sum_{i=1}^{M} (a_0 + a_1 x_i + a_2 x_i^2 - y_i)^2$$

To minimize quadratic error with respect to a_0, a_1, a_2 calculate partial derivatives:

$$\frac{\partial F}{\partial a_0} = 2 \sum_{i=1}^{M} (a_0 + a_1 x_i + a_2 x_i^2 - y_i) = 0$$

$$\frac{\partial F}{\partial a_1} = 2 \sum_{i=1}^{M} x_i (a_0 + a_1 x_i + a_2 x_i^2 - y_i) = 0$$

$$\frac{\partial F}{\partial a_2} = 2 \sum_{i=1}^{M} x_i^2 (a_0 + a_1 x_i + a_2 x_i^2 - y_i) = 0$$

$\Rightarrow$ normal equations:

$$a_0 M + a_1 \sum_{i=1}^{M} x_i + a_2 \sum_{i=1}^{M} x_i^2 = \sum_{i=1}^{M} y_i$$

$$a_0 \sum_{i=1}^{M} x_i + a_1 \sum_{i=1}^{M} x_i^2 + a_2 \sum_{i=1}^{M} x_i^3 = \sum_{i=1}^{M} x_i y_i$$

$$a_0 \sum_{i=1}^{M} x_i^2 + a_1 \sum_{i=1}^{M} x_i^3 + a_2 \sum_{i=1}^{M} x_i^4 = \sum_{i=1}^{M} x_i^2 y_i$$

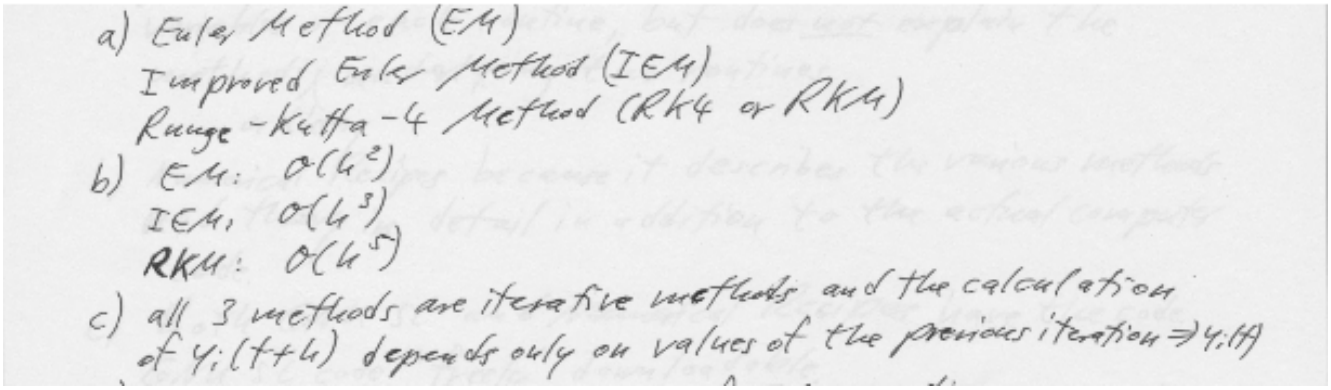
Question #3 (9 points total):

a) List by name three numerical methods covered in class for solving differential equations.
(3 points)

b) List the respective orders of the leading error terms of the three numerical methods in a).
(3 points)

c) When calculating the next segment of the trajectory $y_i(t+h)$ (i.e., time development) of a system, what must all three methods in a) depend on?
(1 point)

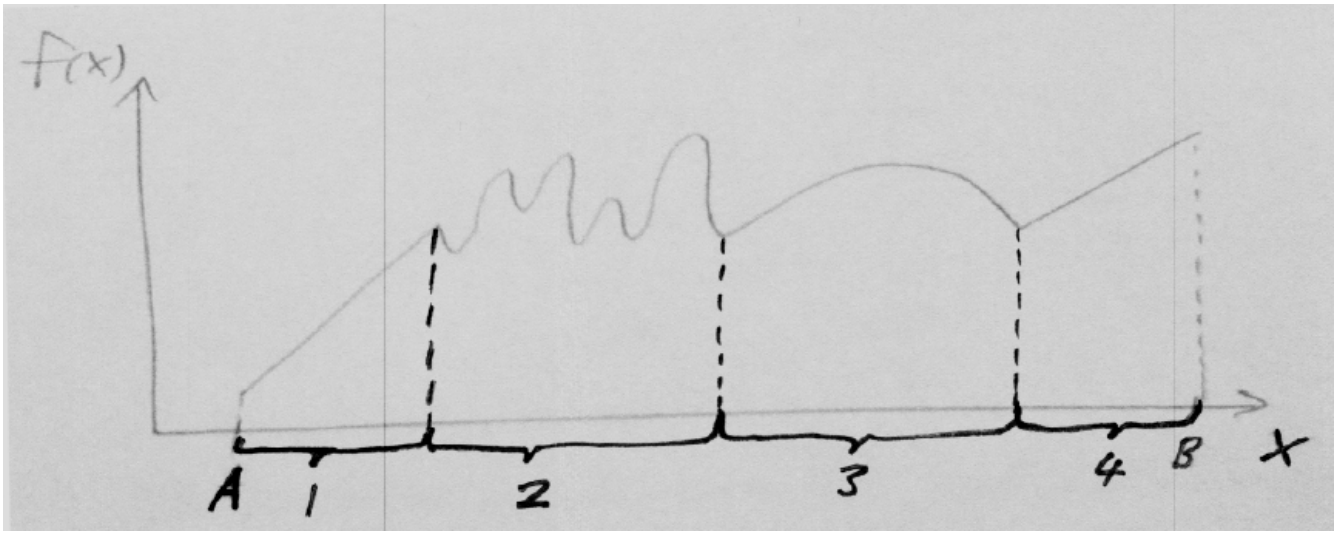
d) When integrating the trajectory of a dynamic system $y(t)$ with the Runge-Kutta-4 method for the interval [0, 1000 sec], how beneficial would it be to subdivide the interval into 100 10 sec intervals ([0, 10 sec], [10 sec, 20 sec], [20 sec, 30 sec], etc.) to obtain a higher overall accuracy for $y(t)$? Justify your answer.
(2 points)



d) A subdivision of the time interval will have no effect, because the correct initial conditions of the dynamic systems are only known for t=0. Therefore, there will be no improvement of the overall accuracy.

Question #4 (4 points total):

How would you integrate f(x) (see graph below) over the interval [A,B]? What technique(s) would you use to get the most accurate numerical results for the integral? Where would you apply the technique(s) in [A,B] and why would it/they be numerically the most accurate?
(4 points)



Region 1: Trapezoidal Rule, because it will most accurately put a straight line across (potentially exact!)

Region 2: Gauss-Legendre Integration with an appropriate number of support points (i.e., high enough degree of Legendre polynomial) will sufficiently approximate this oscillatory segment of the function f(x) (potentially exact if polynomial!)

Region 3: Simpson Rule will place an approximating parabola over this parabolic-looking segment of f(x) (potentially exact!)

Region 4: Trapezoidal Rules will most accurately put a straight line across (potentially exact!)

Question #5 (11 points total):

a) List Rodrigues formula for Legendre Polynomials.
(1 point)

b) Calculate explicity $L_4(x)$ using the Rodrigues formula.
(2 points)

c) Determine the coefficients $C_0, \ldots, C_3$ for a Legendre polynomial expansion of the following polynomial: $P_3(x) = -x^3 - x^2 - x + 1$.
(4 points)

d) Given $P_n(x)$, a polynomial of degree n, and $P_m(x)$, a polynomial of degree m, with n>m, calculate the following expression: $\langle P_n(x) \mid P_m(x) \rangle$. Simplify the result as much as possible.
(4 points)

a) $L_4(x) = \frac{1}{2^4 \cdot 4!} \frac{d^4}{dx^4} (x^2-1)^4$

b) $L_4(x) = \frac{1}{2^4 \cdot 4!} \frac{d^4}{dx^4} (x^2-1)^4$

$= \frac{1}{16 \cdot 24} \frac{d^4}{dx^4} \left[(x^4-2x^2+1)(x^4-2x^2+1)\right]$

$= \frac{1}{384} \frac{d^4}{dx^4} \left[x^8-2x^6+x^4-2x^6+4x^4-2x^2+x^4-2x^2+1\right]$

$= \frac{1}{384} \frac{d^4}{dx^4} \left[x^8-4x^6+6x^4-4x^2+1\right]$

$= \frac{1}{384} \frac{d^3}{dx^3} \left[8x^7-24x^5+24x^3-8x\right]$

$= \frac{1}{384} \frac{d^2}{dx^2} \left[56x^6-120x^4+72x^2-8\right]$

$= \frac{1}{384} \frac{d}{dx} \left[336x^5-480x^3+144x\right]$

$= \frac{1}{384} \left[1680x^4-1440x^2+144\right]$

$= \frac{1}{8} \left[35x^4-30x^2+3\right]$

c)

c) $L_0 = 1$ $\quad L_2 = \frac{3}{2}x^2 - \frac{1}{2}$
$L_1 = x$ $\quad L_3 = \frac{5}{2}x^3 - \frac{3}{2}x$ } use Rodrigues formula for these or recall

$\Rightarrow c_3 = -\frac{2}{5} \Rightarrow c_3 L_3 = -x^3 + \frac{3}{5}x$

$\Rightarrow P_3 - c_3 L_3 = -x^2 - \frac{8}{5}x + 1$

$\Rightarrow c_2 = -\frac{2}{3} \Rightarrow c_2 L_2 = -x^2 + \frac{1}{3}$

$\Rightarrow \left(-x^2 - \frac{8}{5}x + 1\right) - c_2 L_2 = -\frac{8}{5}x + \frac{2}{3}$

$\Rightarrow c_1 = -\frac{8}{5} \Rightarrow c_1 L_1 = -\frac{8}{5}x$

$\Rightarrow \left(-\frac{8}{5}x + \frac{2}{3}\right) - \left(-\frac{8}{5}x\right) = \frac{2}{3}$

$\Rightarrow c_0 = \frac{2}{3}$

$\Rightarrow P_3 = \frac{2}{3}L_0 - \frac{8}{5}L_1 - \frac{2}{3}L_2 - \frac{2}{5}L_3$

d)

b) $P_n(x) = \sum_{i=0} \alpha_i L_i(x) \qquad P_m(x) = \sum_{j=0} \beta_j L_j(x)$

with $n > m$, $n, m \in \mathbb{N}$

$\Rightarrow \langle P_n(x) | P_m(x) \rangle = \langle \sum_{i=0}^{n} \alpha_i L_i(x) | \sum_{j=0}^{m} \beta_j L_j(x) \rangle$

$= \sum_{i=0}^{n} \alpha_i \sum_{j=0}^{m} \beta_j \langle L_i(x) | L_j(x) \rangle$

because of orthogonality $\langle L_i(x) | L_j(x) \rangle = 0$ for $i \neq j$

$= \sum_{k=0}^{m} \alpha_k \beta_k \underbrace{\langle L_k(x) | L_k(x) \rangle}$

$= \sum_{k=0}^{m} \alpha_k \beta_k \underbrace{\frac{2}{2k+1}}$

normalization of $L_k(x)$

$\Rightarrow \boxed{\langle P_n(x) | P_m(x) \rangle = 2 \sum_{k=0}^{m} \frac{\alpha_k \beta_k}{2k+1}}$

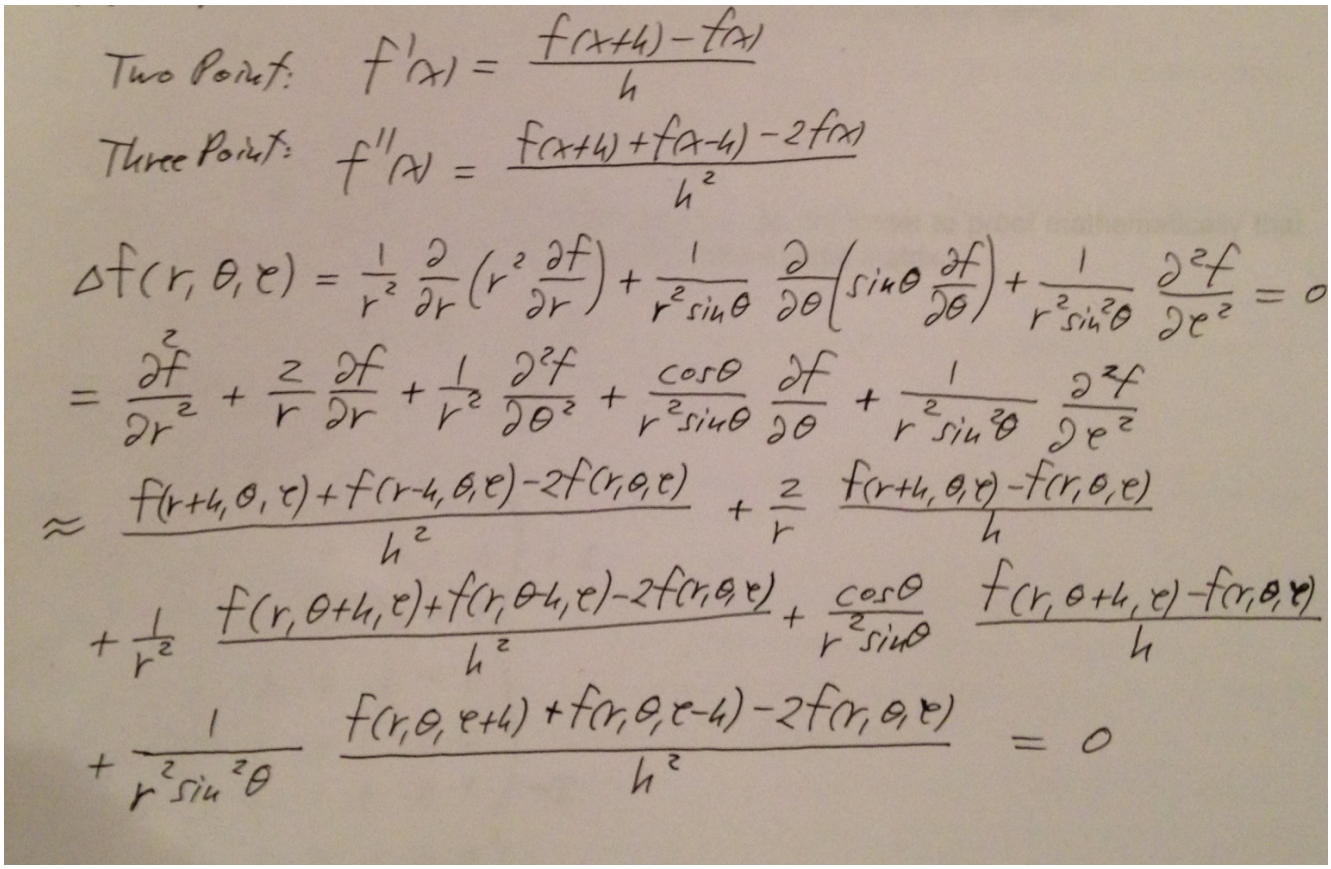
Question #6 (6 points total):

Approximate the general Laplace equation in spherical coordinates with the Two-Point and Three-Point Formulas and ignore the respective leading error terms. Hint: you have to differentiate the brackets first before applying the respective approximation formulas.

The Laplace equation in spherical coordinates is as follows:

$$\Delta f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial f}{\partial \theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial \varphi^2}$$

(6 points)



Two Point: $f'(x) = \frac{f(x+h)-f(x)}{h}$

Three Point: $f''(x) = \frac{f(x+h)+f(x-h)-2f(x)}{h^2}$

$$\Delta f(r,\theta,\varphi) = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial f}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial\varphi^2} = 0$$

$$= \frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r} + \frac{1}{r^2}\frac{\partial^2 f}{\partial\theta^2} + \frac{\cos\theta}{r^2\sin\theta}\frac{\partial f}{\partial\theta} + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial\varphi^2}$$

$$\approx \frac{f(r+h,\theta,\varphi)+f(r-h,\theta,\varphi)-2f(r,\theta,\varphi)}{h^2} + \frac{2}{r}\frac{f(r+h,\theta,\varphi)-f(r,\theta,\varphi)}{h}$$

$$+ \frac{1}{r^2}\frac{f(r,\theta+h,\varphi)+f(r,\theta-h,\varphi)-2f(r,\theta,\varphi)}{h^2} + \frac{\cos\theta}{r^2\sin\theta}\frac{f(r,\theta+h,\varphi)-f(r,\theta,\varphi)}{h}$$

$$+ \frac{1}{r^2\sin^2\theta}\frac{f(r,\theta,\varphi+h)+f(r,\theta,\varphi-h)-2f(r,\theta,\varphi)}{h^2} = 0$$

Question #7 (6 points total):

a) Write down mathematically the Gauss-Legendre Integration (GLI) approximation of an integral of a function f(x) over an arbitrary finite interval [a, b]. Label all components. (3 points)

b) If you had to integrate a polynomial P(x) of degree 330, what would be the highest degree of Legendre polynomial possible to still get an inexact integration result? (1 point)

c) Will there be a difference in the Gauss-Legendre integration of a polynomial P(x) of degree 414 performed with the roots of Legendre polynomial $L_{208}(x)$ versus the roots of $L_{209}(x)$? Explain your answer. (2 points)

a)

$$\int_a^b f(u)\,du = \frac{b-a}{2}\int_{-1}^{1} f\left(\frac{(b-a)x}{2} + \frac{b+a}{2}\right)dx \approx \frac{b-a}{2}\sum_{i=1}^{n} w_i f\left[\frac{(b-a)x_i}{2} + \frac{b+a}{2}\right]$$

(GLI)

with $x_{i=1,\dots n}$ = n roots of Legendre-polynomial L_n in (-1, 1)

and w_i = integration weights

b) L_n integrates polynomials up to P_{2n-1} exactly
==> 2n – 1 = 330 ==> n = 165 (note: n = 166 would yield an exact result)

c) $L_{208} \rightarrow$ 2n – 1 = 415 ==> exact integration
$L_{209} \rightarrow$ 2n – 1 = 417 ==> exact integration

Answer: no, there will be no difference in the integration result of P(x)

Question #8 (4 points total):

a) What does the GNU Scientific Library exactly provide? Give a brief explanation for your answer.
(2 points)

b) If you had a mathematical/computational problem for which you did not know which computational technique/method to use: Would you read up about the methods available in the GNU Scientific Library book or in the Numerical Recipes book? Give a brief explanation for your answer.
(2 points)

a) GNU SL is a library of numerical routines for solving numerical problems. GNU SL describes/lists the usage and parameters/variables of each routine, but does not explain the methods underlying the routines.

b) NR because it describes and derives the various methods in detail in addition to discussing the actual computer code.